```python
# Implementation of Hierarchical Clustering using Agglomerative
Approach
# and Visualization using Dendrogram

import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.datasets import make_blobs
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import AgglomerativeClustering
from scipy.cluster.hierarchy import dendrogram, linkage
from sklearn.metrics import silhouette_score

print("SHRAWANI KHANDAGALE")
print("Roll No.-69")
# 1. Generate Dataset
X, _ = make_blobs(n_samples=100, centers=3, random_state=42)

# 2. Standardize the Data
X_scaled = StandardScaler().fit_transform(X)

# 3. Create and Display Dendrogram
linked = linkage(X_scaled, method='ward')

plt.figure(figsize=(10, 5))
dendrogram(linked)
plt.title("Dendrogram - Hierarchical Clustering")
plt.xlabel("Data Points")
plt.ylabel("Euclidean Distance")
plt.show()

# 4. Apply Agglomerative Clustering
model = AgglomerativeClustering(n_clusters=3, linkage='ward')
clusters = model.fit_predict(X_scaled)

# 5. Display Clustered Data
plt.figure(figsize=(8, 5))
sns.scatterplot(x=X_scaled[:, 0], y=X_scaled[:, 1],
                hue=clusters, palette='viridis', s=60)
plt.title("Agglomerative Hierarchical Clustering")
plt.xlabel("Feature 1 (Scaled)")
plt.ylabel("Feature 2 (Scaled)")
plt.legend(title="Cluster")
plt.show()

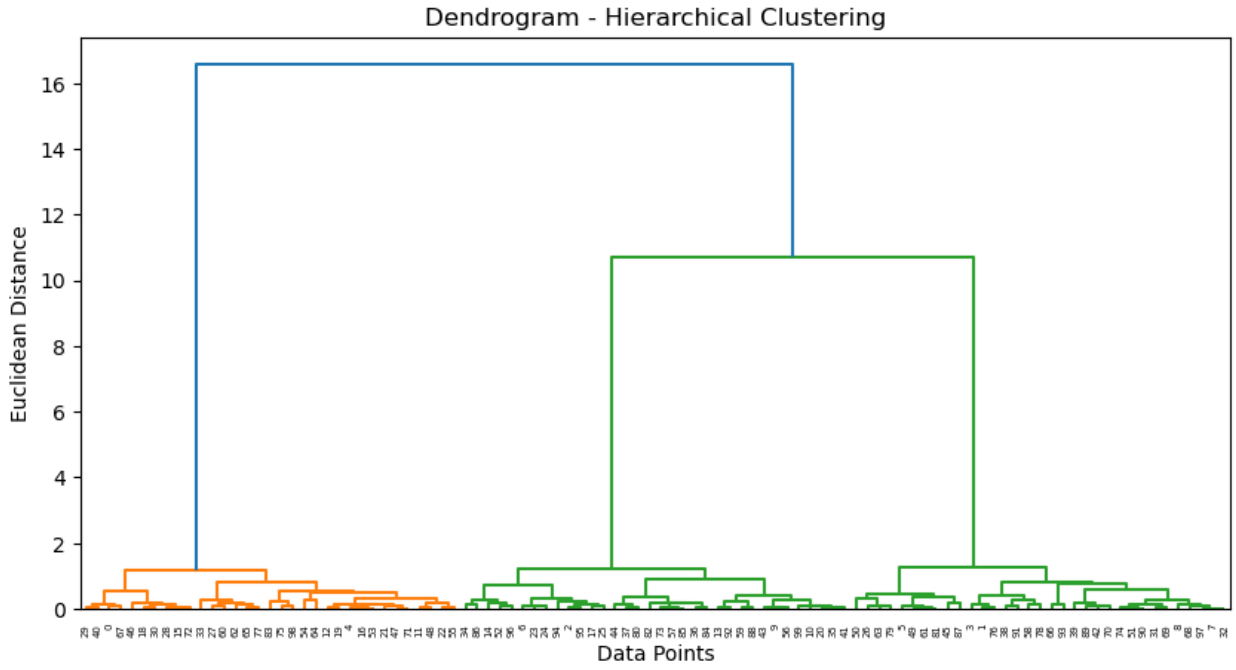
# 6. Display Cluster Information
print("Cluster Labels:")
print(clusters)

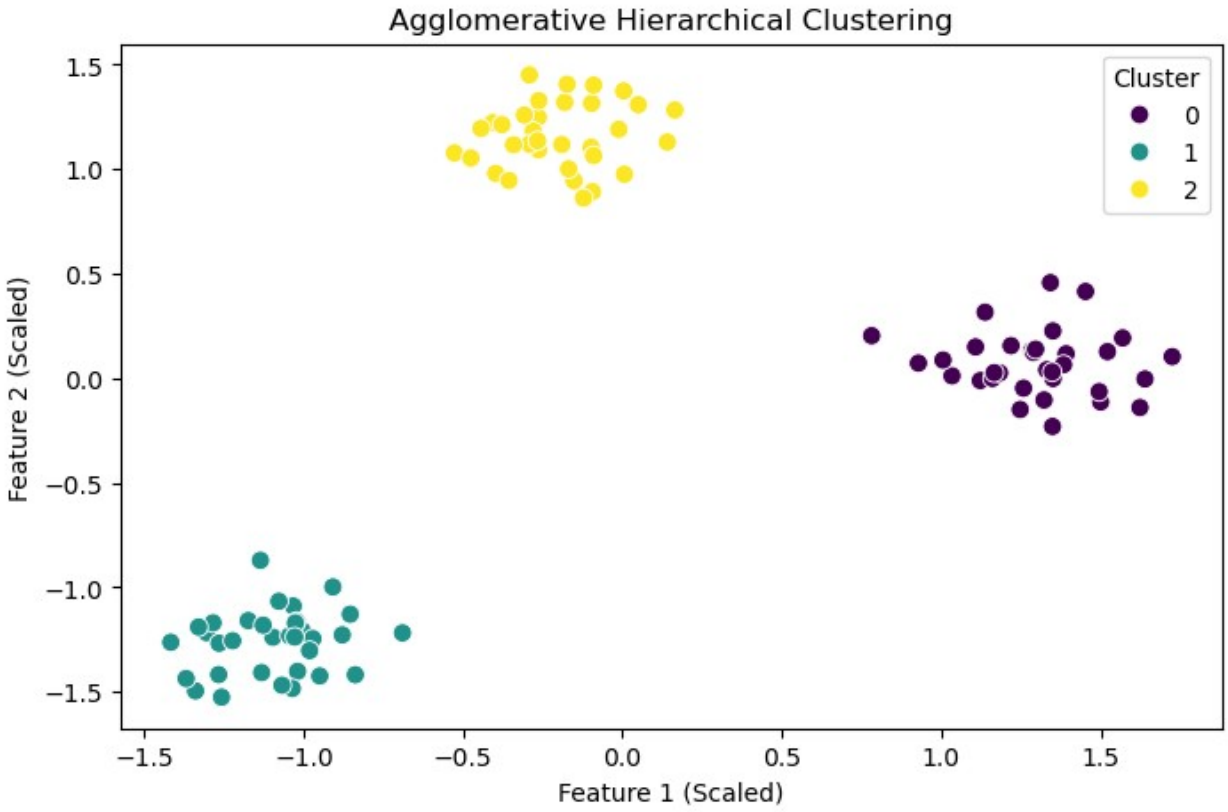
print("\nNumber of clusters:", len(set(clusters)))
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# 7. Calculate Silhouette Score
score = silhouette_score(X_scaled, clusters)
print("Silhouette Score:", round(score, 3))

print("\nConclusion:")
print("Agglomerative clustering combines nearby data points into
clusters.")
print("The dendrogram shows how clusters merge at different
distances.")
print("The scatter plot displays the final three clusters.")

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```






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Cluster Labels:
[1 0 2 0 1 0 2 0 0 2 2 1 1 2 2 1 1 2 1 1 2 1 1 2 2 2 0 1 1 1 1 0 0 1 2
2 2
 2 0 0 1 2 0 2 2 0 1 1 1 0 0 0 2 1 1 1 2 2 0 2 1 0 1 0 1 1 0 1 0 0 0 1
1 2
 0 1 0 1 0 0 2 0 2 1 2 2 2 0 2 0 0 0 2 0 2 2 2 0 1 2]

Number of clusters: 3
Silhouette Score: 0.849

Conclusion:
Agglomerative clustering combines nearby data points into clusters.
The dendrogram shows how clusters merge at different distances.
The scatter plot displays the final three clusters.
```